

Study details

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Abstract

We synthesized the first multifunctionalized phosphoinositide polyphosphate derivatives featuring a photo-removable protecting group (“cage”), a photo-crosslinkable diazirine group, and a terminal alkyne group useful for click chemistry. We demonstrate that the lipid derivatives readily enter cells. After photo-crosslinking, cell fixation and fluorescent tagging via click chemistry, we determined the intracellular location of the lipid derivatives before and after uncaging of the lipids. We find that there is rapid trafficking of PI(3,4)P2 and PI(3,4,5)P3 derivatives to the plasma membrane, opening the intriguing possibility that there is active transport of these lipids involved. We employed the photo-crosslinking and click chemistry functions to analyze the proteome of PI(3,4,5)P3-binding proteins. From the latter, we validated by RNAi that the putative lipid binding proteins ATP11A and MPP6 are involved in the transport of PI(3,4,5)P3 to the plasma membrane.

Lipid probes utilized

[Trifunctional PI\(3,4,5\)P₃](#)

Cell line analyzed

[HeLa](#)

Species: *Homo sapiens*

Cell type: Epithelial (immortalized cervical cancer line)

Mass spectrometry quantification method

Peptide spectral counting (*in original publication*)

Label free quantification (*as reported here, enabled via fresh analysis by Frank Stein, EMBL – details below*)

Additional sample preparation ?

Samples were split into membrane and cytosolic fractions and analyzed as independent mass spec samples. Additional *No Probe* controls were included for raw input, though not used in displaying the data on this site.

Data analysis

The authors of this paper were unable to provide the original results of their dataset, and we applied a label-free quantification data pipeline to their mass spectrometry files, publicly-available on ProteomExchange

For this analysis, raw protein intensities were processed by first selecting unique proteins supported by at least two razor peptides. Batch effects were then removed, and missing values were imputed using a hybrid strategy: KNN imputation was applied to high-confidence proteins with replicate-level missingness, while low-confidence proteins were assigned minimum detected values (MinDet). The data were subsequently variance-stabilized and normalized, after which differential abundance testing was performed using the limma framework. Parallel analyses were also carried out with the fdrtool package, with significance thresholds defined consistently across methods. Proteins were classified according to standard criteria: hits ($\text{FDR} < 0.05$ and $|\text{fold-change}| > 1.5$), candidates ($\text{FDR} < 0.2$ and $|\text{fold-change}| > 1.5$), enriched candidates ($\text{FDR} < 0.2$ and $\text{fold-change} > 1.5$), enriched hits ($\text{FDR} < 0.05$ and $\text{fold-change} > 1.5$), and all others as non-significant.

Data wrangling

Use the download button below to download the R script used to wrangle the authors' original submission into the data visualization tools on the Lipid Interactome:

Use the download button below to download the original data file prior to wrangling: